

Repair-Priority Guided Endpoint Flow Postfilter for Noise-Robust Neural Speech Codecs

Pincheng Lu, Xiaojiao Chen, Jing Wang, Peng Zhou, Zhihang Cheng, and Shenghui Zhao

Abstract—Noise-robust neural speech codecs often suffer from perceptual degradation at low bitrates and low signal-to-noise ratios. While generative postfilters can improve quality, they typically perform uniform global restoration, ignoring perceptually critical regions that remain severely distorted. In this letter, we propose FlowRep, a repair-priority guided endpoint flow postfilter for noise-robust speech codecs. Specifically, a Repair-Priority Estimator predicts a priority map to reweight the training objective, guiding the flow model to focus on regions requiring critical restoration. Furthermore, we formulate post-enhancement as endpoint flow matching. Through direct prediction of the clean spectral target from intermediate states, this formulation naturally accommodates metric-oriented objectives and provides a flexible quality-complexity trade-off via adjustable inference steps. Experimental results demonstrate that our method significantly improves the perceptual quality and intelligibility of codec-decoded speech, especially under challenging low-SNR conditions.

Index Terms—Flow matching, speech codecs, speech enhancement, postfiltering

I. INTRODUCTION

Recent neural speech codecs (e.g., EnCodec [1], DAC [2]) excel in clean-speech reconstruction but lack robustness in noisy environments. To address this, codecs with integrated enhancement, such as SoundStream [3], SEStream [4], and NoiseRobustVRVQ [5], have been developed. However, their performance remains highly sensitive to the input signal-to-noise ratio (SNR); at low SNRs, the decoded speech still suffers from severe detail loss and spectral distortion.

To further improve decoded quality, generative postfilters like ScoreDec [6] and FlowDec [7] employ diffusion- or flow-based models to restore details lost during compression. Nevertheless, these methods are primarily designed for clean speech. For Noise-Robust Codecs, the residual degradations are caused by the joint effects of lossy compression and noise suppression, making them highly non-uniform across time-frequency regions. Existing postfilters typically perform uniform global restoration, lacking the targeted focus needed to efficiently repair these localized, perceptually critical errors. This motivates the need for a codec-aware postfilter capable of priority-guided selective restoration.

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The authors are with the Beijing Institute of Technology, Beijing 100081, China (e-mail: parslog@163.com; xiaojiaoch@163.com; wangjing@bit.edu.cn; p.zhou@bit.edu.cn; czhhcheng@163.com; shzhao@bit.edu.cn).

In this letter, we propose **FlowRep**, a repair-priority guided endpoint flow postfilter tailored for end-to-end speech enhancement and compression. By combining a Repair-Priority Estimator (RPE) for targeted restoration with an endpoint flow matching formulation for direct clean-speech prediction, FlowRep efficiently repairs non-uniform degradations while maintaining a stable, metric-compatible training paradigm.

Our main contributions in this work are as follows: (1) we propose a Repair-Priority Estimator to predict a Repair-Priority Map, guiding the postfilter to focus on severely distorted regions; (2) we develop an endpoint flow matching postfilter that enables metric-specific optimization and reduces the Number of Function Evaluations (NFE) for high-quality generation.

II. METHOD

A. Overall Framework

The overall framework of the proposed FlowRep is illustrated in Fig. 1. Given a noisy waveform $\mathbf{x}_n \in \mathbb{R}^T$, we extract its complex spectrogram $\mathbf{S}_n \in \mathbb{C}^{F \times T'}$ via short-time Fourier transform (STFT). A *Noise-Robust Codec* processes $\mathbf{S}_n$ for joint enhancement and compression, yielding a relatively clean decoded spectrogram $\mathbf{S}_d$. Next, utilizing $\mathbf{S}_d$, decoder features, and encoder-side cues, a *Repair-Priority Estimator* generates a binary map $\mathbf{M} \in \{0, 1\}^{F \times T'}$, where 1 indicates regions needing focused restoration and 0 denotes mild refinement. Finally, guided by $\mathbf{M}$, an *endpoint flow matching* postfilter iteratively refines the $\mathbf{S}_d$ into a high-fidelity speech spectrogram (see Fig. 2 for detailed visualizations).

B. Noise-Robust Codec

We employ *FreqCodec* [8] as the backbone for joint enhancement and compression on complex spectrograms. Its encoder uses four cascaded residual 2-D convolutional blocks to extract continuous latents. These latents are discretized by a finite scalar quantizer (FSQ) [9], which ensures stable training by avoiding learnable codebooks and associated commitment losses. Finally, a mirrored decoder reconstructs the quantized latents into $\mathbf{S}_d$. To achieve noise robustness, the entire codec is trained on paired noisy-clean speech.

C. Repair-Priority Estimator

Inspired by the concept of importance maps in image compression [10]–[13] and neural speech coding [14], the *Repair-Priority Estimator (RPE)* identifies time-frequency regions requiring targeted restoration. It takes three inputs: the decoded

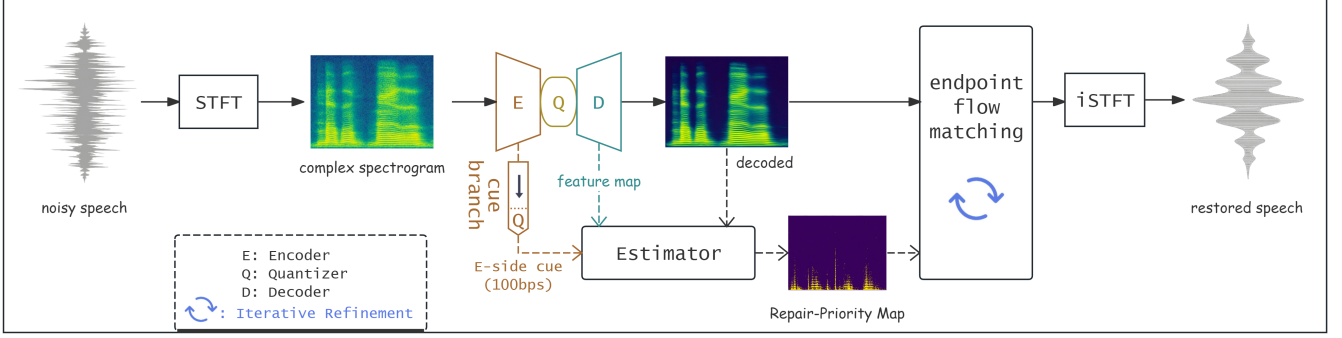


Fig. 1. Overall framework of the proposed **FlowRep**. The **circular arrow** denotes **iterative refinement at inference time**, where the endpoint flow matching postfilter supports adjustable **NFE** for progressive quality enhancement.

spectrogram $\mathbf{S}_d$, decoder features $\mathbf{F}_d$, and an encoder-side cue $\hat{\mathbf{Z}}_s$. Specifically, the cue branch extracts compact pre-quantization information from encoder features $\mathbf{F}_e$ to provide source-aware cues that facilitate the prediction of $\mathbf{M}$. This representation is discretized via FSQ as $\hat{\mathbf{Z}}_s$ and transmitted at 100 bps.

The inputs are projected by embedding functions $\phi(\cdot)$ and fused via channel-wise concatenation into a representation $\mathbf{H} = [\phi_{\text{spec}}(\mathbf{S}_d), \phi_{\text{feat}}(\mathbf{F}_d), \phi_{\text{side}}(\hat{\mathbf{Z}}_s)]$. An NCSN++ [15] backbone processes $\mathbf{H}$ to predict a normalized continuous map $\tilde{\mathbf{M}} \in (0, 1)^{F \times T'}$ via a sigmoid function, which is then binarized into $\mathbf{M} \in \{0, 1\}^{F \times T'}$ using a threshold of 0.5. Here, $\mathbf{M} = 1$ indicates regions needing focused restoration.

During the joint training of the RPE and the cue branch, we utilize paired clean and noisy spectrograms $(\mathbf{S}_c, \mathbf{S}_n)$. The noisy input $\mathbf{S}_n$ is first processed by the Noise-Robust Codec to yield the decoded spectrogram $\mathbf{S}_d$. Since the binarization of the priority map is non-differentiable, we apply the straight-through estimator (STE) during back-propagation:

$$\mathbf{M}_{\text{STE}} = \text{stopgrad}(\mathbf{M} - \tilde{\mathbf{M}}) + \tilde{\mathbf{M}}. \quad (1)$$

Subsequently, a hybrid magnitude spectrogram $|\mathbf{S}_h|$ is constructed by combining the codec-degraded representation $|\mathbf{S}_d|$ and the clean target $|\mathbf{S}_c|$ according to the estimated priority map:

$$|\mathbf{S}_h| = (\mathbf{1} - \mathbf{M}_{\text{STE}}) \odot |\mathbf{S}_d| + \mathbf{M}_{\text{STE}} \odot |\mathbf{S}_c|, \quad (2)$$

where $\odot$ denotes element-wise multiplication.

We employ an adversarial training strategy to help the RPE locate severely distorted regions. A discriminator D_η is trained via a least-squares objective to distinguish $|\mathbf{S}_h|$ from the clean target $|\mathbf{S}_c|$, while the RPE learns to fool D_η by prioritizing the most degraded bins. To prevent the RPE from trivially selecting the entire spectrogram, we apply a sparsity constraint on $\tilde{\mathbf{M}}$. The overall objective combines the adversarial loss and sparsity penalty:

$$\mathcal{L}_D = \mathbb{E} \left[(1 - D_\eta(|\mathbf{S}_c|))^2 \right] + \mathbb{E} \left[(D_\eta(|\mathbf{S}_h|))^2 \right], \quad (3)$$

$$\mathcal{L}_{\text{RPE}} = \mathbb{E} \left[(1 - D_\eta(|\mathbf{S}_h|))^2 \right] + \lambda_{\text{sp}} \text{mean}(\tilde{\mathbf{M}}), \quad (4)$$

where λ_{sp} controls the sparsity strength. Driven by this objective, the cue branch implicitly learns to transmit task-oriented pre-quantization information without explicit supervision.

D. Endpoint Flow Postfilter

Inspired by FlowDec [7], we employ a flow matching postfilter to repair the decoded speech. To overcome the ambiguous velocity estimation of traditional methods in silent or low-energy regions, we adopt an endpoint-supervised formulation [16]. Specifically, the network g_θ is trained to directly predict the clean ground-truth endpoint $\mathbf{S}_1$ from an intermediate state $\mathbf{S}_t$:

$$\hat{\mathbf{S}}_1 = g_\theta(\mathbf{S}_t, t \mid \mathbf{c}, \mathbf{M}). \quad (5)$$

This formulation provides three key advantages: **1) Stable low-variance supervision** via a fixed, acoustically meaningful target; **2) Constraint-friendly optimization** that allows direct application of perceptual metrics on the output; and **3) Efficient few-step sampling** by encouraging a direct restoration trajectory. A formalized mathematical analysis of these advantages is provided in the supplementary material.

In the forward process, the initial state $\mathbf{S}_0$ is sampled from a conditional Gaussian centered at the decoded speech $\mathbf{S}_d$, which also serves as the condition $\mathbf{c} = \mathbf{S}_d$. The intermediate state $\mathbf{S}_t$ is obtained via linear interpolation:

$$\mathbf{S}_0 = \mathbf{S}_d + \epsilon, \quad \mathbf{S}_t = (1 - t)\mathbf{S}_0 + t\mathbf{S}_1, \quad (6)$$

where ϵ is a complex Gaussian noise matrix with i.i.d. elements drawn from $\mathcal{CN}(0, \sigma^2)$, and $t \sim \mathcal{U}(0, 1)$.

To focus the postfilter on critical regions, we reweight the flow matching objective using the weighting map $\mathbf{M} + 1$, which assigns a weight of 2 to distorted regions ($\mathbf{M} = 1$) and 1 otherwise ($\mathbf{M} = 0$). Here, $|\cdot|^2$ denotes the squared magnitude of the complex error, yielding an $F \times T'$ time-frequency error map that matches the dimensions of $\mathbf{M}$. The repair-priority weighted endpoint flow matching loss is formulated as:

$$\mathcal{L}_{\text{FM}} = \mathbb{E}_{t, \mathbf{S}_0, \mathbf{S}_1} \left[\text{mean} \left((\mathbf{M} + 1) \odot |\hat{\mathbf{S}}_1 - \mathbf{S}_1|^2 \right) \right]. \quad (7)$$

Furthermore, to impose perceptually motivated constraints, we apply a differentiable PESQ surrogate loss¹ between the time-domain output $\hat{\mathbf{x}}_1$ (obtained via ISTFT) and the clean reference $\mathbf{x}_1$. Minimizing this surrogate encourages improvements in the PESQ-related perceptual quality. The overall training objective is:

$$\mathcal{L}_{\text{post}} = \mathcal{L}_{\text{FM}} + \lambda_{\text{PESQ}} \mathcal{L}_{\text{PESQ}}(\hat{\mathbf{x}}_1, \mathbf{x}_1), \quad (8)$$

¹<https://github.com/audiolabs/torch-pesq>

where λ_{PESQ} is a balancing hyperparameter.

During inference, we generate the enhanced speech by integrating the learned deterministic flow starting from $\mathbf{S}_0$. Since the network predicts the clean endpoint, the velocity field is derived as

$$\mathbf{v}_\theta = (\hat{\mathbf{S}}_1 - \mathbf{S}_t)/(1 - t). \quad (9)$$

The final enhanced spectrogram is obtained as:

$$\hat{\mathbf{S}} = \mathbf{S}_0 + \int_0^1 \frac{g_\theta(\mathbf{S}_t, t \mid \mathbf{c}, \mathbf{M}) - \mathbf{S}_t}{1 - t} dt. \quad (10)$$

In practice, to avoid numerical singularities at the boundaries (due to the log-scaled time embedding at $t = 0$ and the velocity denominator at $t = 1$), the integration is performed over a truncated interval $[\delta, 1 - \delta]$ for a small constant $\delta > 0$.

III. EXPERIMENT

A. Model Configuration

The input speech is processed via STFT (FFT size 512, window length 480, hop length 160). The encoder consists of four 128-channel cascaded blocks with temporal downsampling factors of (2, 1, 1, 1) and no frequency downsampling. Its output is projected to 4 frequency bins and flattened, yielding a 50 Hz main latent representation for 16 kHz speech. The decoder mirrors the encoder’s architecture.

The encoder-side cue branch takes the channel-wise concatenation of the output feature maps from all encoder blocks as input. It further downsamples the temporal dimension by a factor of 5 and projects the frequency dimension to 4 bins, producing a compact cue representation at 10 Hz. This branch is quantized via FSQ and assigned a fixed bitrate of 100 bps, which is included in the total reported bitrate.

We evaluate our model at two total bitrate settings: 1 kbps and 6 kbps. The detailed bitrate allocations and FSQ configurations ($\mathbf{L}$) for both the main codec stream and the cue branch are summarized in Table I. Fully connected layers are employed before and after quantization to handle dimension matching.

TABLE I
BITRATE ALLOCATION AND FSQ CONFIGURATIONS

Total	Main (bps)	$\mathbf{L}_{\text{main}}$	Cue (bps)	$\mathbf{L}_{\text{cue}}$
1 kbps	900	[8, 8, 8, 8, 8, 8]	100	[4, 4, 4, 4, 4]
6 kbps	5900	$\{[8, 8, 8, 8, 8, 8] \times 6, [4, 4, 4, 4, 4, 4]\}$	100	[4, 4, 4, 4, 4]

B. Datasets

We used a combination of multiple speech datasets for training and evaluation. For **clean speech data**, we employed LibriTTS [17], VCTK [18], and AISHELL-3 [19]. For **paired noisy-clean speech data**, we used VoiceBank+DEMAND [20] and DNS-Challenge [21], where the DNS-Challenge data covers SNRs from -5 to 20 dB. All audio was downsampled to 16 kHz. To rigorously assess the postfilter’s robustness, the noisy evaluation subset is strictly sampled from challenging low-SNR conditions from -5 to 5 dB.

C. Training and Evaluation Settings

Training and Inference Settings: The system is trained in three stages. First, the noise-robust FreqCodec is trained for 250k iterations using both high-quality clean speech and paired noisy-clean speech. With the codec frozen, the RPE and the cue branch are trained for 50k iterations. Finally, the endpoint flow matching postfilter is trained for another 50k iterations. The loss weights are set to $\lambda_{\text{sp}} = 10$ and $\lambda_{\text{PESQ}} = 1$, and the standard deviation of the initial noise is set to $\sigma = 0.487$. During inference, we use the Euler ODE solver, restricting the integration time to $[\delta, 1 - \delta]$ with $\delta = 0.03$.

Evaluation Metrics: We constructed the evaluation set with 200 clean utterances and 200 noisy-clean utterance pairs for objective evaluation, and randomly selected 20 clean utterances and 20 noisy-clean utterance pairs for subjective evaluation.

For objective evaluation, we report PESQ [22] to measure perceptual quality distortion and compute WER using Whisper-tiny [23] to assess intelligibility distortion. For subjective evaluation, 10 listeners, including 5 females and 5 males aged between 22 and 29, participated in the listening tests. We conduct MUSHRA tests [24] for both clean and noisy speech evaluation, and report the mean scores together with 95% confidence intervals.

Baselines: We compare the proposed method with four representative baselines:

- **FreqCodec** [8]: the underlying codec without postfiltering.
- **FreqCodec + One-step NCSN++** [15]: an NCSN++-based one-step postfilter.
- **FreqCodec + ScoreDec-style SPF** [6]: an adapted score-based diffusion postfilter conditioned on the FreqCodec output.
- **FreqCodec + FlowDec-style CFM** [7]: an adapted conditional flow matching postfilter following FlowDec.

For fair comparison, all postfilter baselines use the same FreqCodec decoded speech as the condition and are trained on the same data as our method. All postfilters are implemented with the same NCSN++ backbone and comparable model capacity, differing only in the input layer and their corresponding training and generation formulations. For ease of notation, in the following sections, NCSN++, ScoreDec, FlowDec, and FlowRep denote the same FreqCodec backbone cascaded with the corresponding postfilter.

D. Results Analysis

As shown in Table II and Fig. 3, FlowRep consistently outperforms all baselines across all objective and subjective metrics. This comprehensive improvement can be attributed to the combination of endpoint flow matching, repair-priority guidance, and metric-oriented training. Through direct prediction of the clean spectral endpoint, this formulation inherently supports high-quality generation with a significantly smaller NFE. Moreover, it naturally accommodates a differentiable PESQ loss, explicitly boosting the objective perceptual metric. Furthermore, this explicit target formulation faithfully restores the original spectral structure and effectively mitigates the

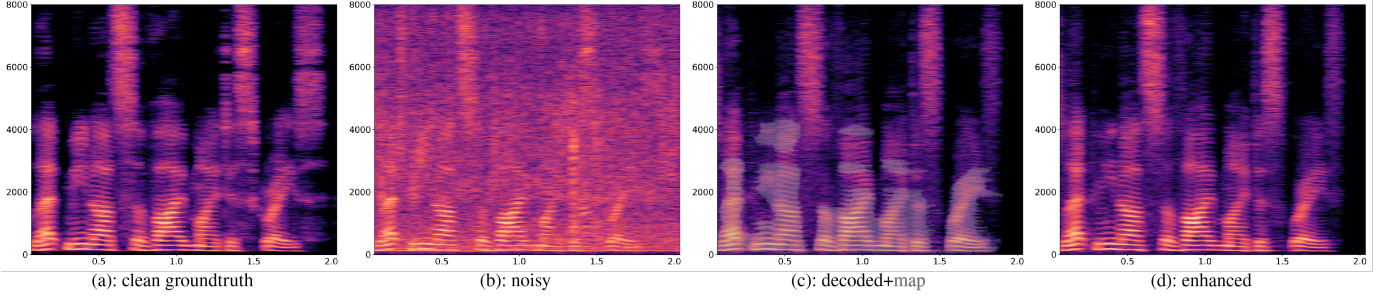


Fig. 2. Speech spectrograms before and after coding and enhancement. The repair-priority map in (c) is visualized by overlaying a semi-transparent mask on the decoded spectrogram.

TABLE II
OBJECTIVE EVALUATION RESULTS UNDER CLEAN AND NOISY CONDITIONS.

Model	Bitrate (bps)	NFE	Noisy		Clean	
			PESQ $\uparrow$	WER($\%$) $\downarrow$	PESQ $\uparrow$	WER($\%$) $\downarrow$
FreqCodec	1k	–	1.5953	55.48	2.2157	13.84
NCSN++	1k	–	1.6438	55.29	2.2665	13.66
ScoreDec	1k	50	1.7577	52.73	2.3257	12.88
FlowDec	1k	6	1.8165	49.32	2.3975	12.56
FlowRep	1k	3	1.9164	39.21	2.5785	9.57
FreqCodec	6k	–	2.0457	41.41	3.9276	3.23
NCSN++	6k	–	2.0928	39.88	4.0255	3.09
ScoreDec	6k	50	2.1332	37.14	4.1360	2.97
FlowDec	6k	6	2.1893	36.16	4.1875	2.88
FlowRep	6k	3	2.3535	30.75	4.2978	1.06

TABLE III
ABLATION RESULTS UNDER THE NOISY CONDITION AT 6 KBPS.

Model	PESQ $\uparrow$	WER($\%$) $\downarrow$
FlowRep@6kbps (NFE=3)	2.3535	30.75
w/o RPE	2.3069	32.68
w/o $\mathcal{L}_{\text{PESQ}}$	2.2312	30.86
– w/o $\mathcal{L}_{\text{PESQ}}$ and endpoint supervision	2.1247	40.49

accumulated distortions caused by ambiguous velocity estimation. This reduction in structural distortion ensures high speech intelligibility, leading to a lower WER. Finally, since human auditory perception relies heavily on accurate spectral patterns, such faithful reconstruction directly translates to the superior subjective quality observed in the MUSHRA tests.

Furthermore, the RPE plays a crucial role in handling non-uniform distortions. As visualized in Fig. 2(c), the generated RPE map accurately covers the dense low-frequency regions rich in formants, as well as specific high-frequency areas containing critical content. These are precisely the regions most vulnerable to over-smoothing and structural distortion during the base codec’s processing. By explicitly highlighting these distortion-prone areas, the map effectively guides the subsequent flow matching model to concentrate its modeling capacity where it is needed most. As a result, the postfilter avoids redundant global processing and successfully repairs the targeted regions, restoring sharp and faithful spectral details (Fig. 2(d)).

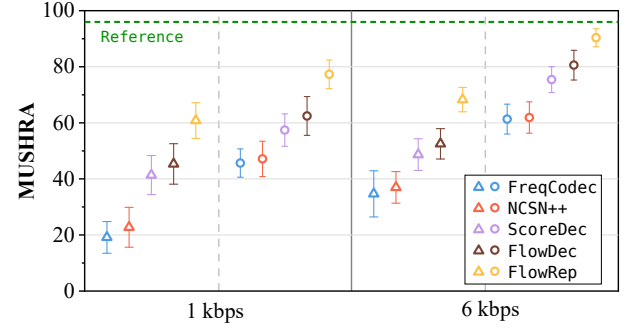


Fig. 3. Subjective MUSHRA scores for noisy and clean speech. $\triangle$ denotes noisy speech, while $\circ$ denotes clean speech. Colors indicate different post-filtering methods: FreqCodec, NCSN++, ScoreDec, FlowDec and FlowRep.

E. Ablation Study

Table III presents the ablation study of FlowRep’s key components. First, removing the RPE and reallocating its 100 bps to the main codec degrades both PESQ and WER. This confirms that the RPE effectively guides the postfilter to concentrate on distortion-prone regions rather than performing uniform processing.

Next, removing $\mathcal{L}_{\text{PESQ}}$ causes a PESQ drop of about 0.1 with stable WER, demonstrating that metric-specific guidance successfully yields targeted perceptual improvements.

Finally, removing both $\mathcal{L}_{\text{PESQ}}$ and endpoint supervision degrades the model to a standard FlowDec-style baseline. It is worth noting that at NFE=3, this degraded model performs significantly worse than the FlowDec (NFE=6) in Table II. This further confirms the superiority of the endpoint flow matching formulation for few-step speech restoration.

IV. CONCLUSION

In this letter, we propose FlowRep, a repair-priority guided endpoint flow postfilter for noise-robust neural speech codecs. By introducing a Repair-Priority Estimator, the proposed method encourages the postfilter to focus on perceptually critical time-frequency regions after codec decoding. We further formulate post-decoding restoration as endpoint flow matching, which enables direct endpoint supervision with spectral and perceptual objectives. Experimental results demonstrate that FlowRep improves the perceptual quality of codec-decoded speech under both clean and noisy conditions, especially in challenging low-SNR scenarios.

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